

Software

The software was developed using four tools: CodeVisionAVR, Proteus, SinaProg and PCB Wizard.

CodeVisionAVR.

The coding and hex file were generated using the Evaluation version of CodeVisionAVR tool. In CodeVisionAVR, open a new project and select 'Target AVR Chip' as ATmega. Select the ATmega16 chip with 8MHz clock frequency, followed by output port as Port B0. Then select 'ADC Enabled' option and set voltage reference as AREF pin with 1000kHz clock. Next, select 'Enable Alphanumeric LCD Support' option followed by Port C (PC0-RP6) with 16 characters.

In 'Program' menu, select 'Generate, Save and Exit' option. Save the file three times with the same name but different extensions as indicated by the software. In the coding window, start code within the while loop to read the sensor output signal at AD0 pin of the microcontroller. Next, put a condition if reference voltage 2V is greater than or equal to the sensor signal. Then build the entire project file to produce the hex file.

Proteus. Open Proteus simulation tool, create a new file, then open the library and select the required components (Fig. 4). To simulate the code, double-click the controller (U1) and insert the hex file. Simulation runs when the sensed voltage is higher than the reference voltage (2V). The LCD shows that the soil is dry and the motor is 'on.' When the sensed voltage is less than the reference voltage, the LCD shows the soil as 'wet' and the motor as 'off.'

USBasp. USBasp is a USB in-

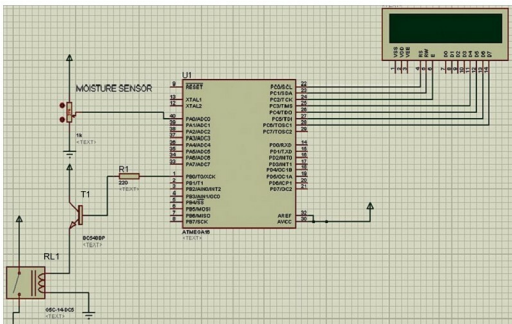


Fig. 4: Proteus simulation without DC motor

circuit programmer for Atmel AVR controllers, similar to AVRPro+ board. It consists of an ATmega88 or an ATmega8 and a few passive components. The programmer uses a firmware-only USB driver, requiring no particular USB controller. It is used along with SinaProg software to burn the hex file into the controller.

SinaProg. SinaProg software is used to burn the hex code and set fuse bits for the AVR microcontroller. It is a good software tool for hobbyists and experimenters alike. Connect the AVR USB programmer such as USBasp to your PC. Load the hex code, select the device programmer and set the fuses (Int.2MHz) as shown in Fig. 7.

PCB Wizard. PCB Wizard is used to design the circuit board. Open the tool and select the required components from the PCB component gallery. (You can also use any other PCB designing tool such as open source software gEDA.)

Construction and testing

An actual-size PCB layout of the automatic drip irrigation system

PARTS LIST

Semiconductors:

- IC1 - ATmega16 microcontroller
- IC2 - 7805, 5V voltage regulator
- LED1 - 5mm LED
- D1 - 1N4007 rectifier diode
- T1 - BC548 npn transistor

Resistors (all 1/4-watt ±5% carbon):

- R1 - 680-ohm
- R2, R4 - 220-ohm
- R3 - 10-kilo-ohm
- VR1 - 10-kilo-ohm preset

Capacitors:

- C1 - 470pF, 25V electrolytic
- C2 - 0.1µF ceramic disk
- C3 - 10µF, 16V electrolytic

Miscellaneous:

- LCD1 - 16x2 LCD module
- CON1, CON2 - 2-pin terminal connector
- CON3 - 3-pin berg strip female connector
- S1 - Tactile switch
- RL1 - 12V, 1C/O relay
- M1 - DC motor aquarium pump
- BATT.1 - 9V PP3 battery
- BATT.2 - 12V battery/12V DC, 1A power supply
- IC1 - 40-pin IC base for IC1
- Heat-sink for IC2
- YL-38 and YL-69 moisture sensor

is shown in Fig. 5 and its components layout in Fig. 6. After assembling the circuit, enclose it in a suitable box.

First, program the microcontroller with the hex code and place